

Minutes of the 8th meeting of the Board of Studies of the Department of Computer Science, held on 15th November, 2023

The 8th meeting of the board of study of the Department of Computer Science, University of Swabi was held on 15th November, 2023, at 10:00 AM in the committee room, Department of Computer Science. The meeting was chaired by Prof. Dr. Mukhtar Alam, Dean Faculty of Science, University of Swabi. The following members attended the meeting:

S.No.	Name of Faculty Member	Designation	University/Institution
1	Prof. Dr. Mukhtar Alam	Dean Faculty of Science	University of Swabi
2	Dr. Shah Nazir	Assistant Professor	University of Swabi
3	Mr. Anwar Hussain	Lecturer	University of Swabi
4	Mr. Yaser Salim	Lecture	University of Swabi
5	Dr. Sara Shahzad	Professor	University of Peshawar
6	Dr. Sajid Anwar	Associate Professor	IMSciences, Peshawar
7	Mr. Inayat Ullah	Lecturer	GDC Lahore, Swabi

Proceedings of the meeting were started with the recitation of the holy Quran by Dr. Shah Nazir, Assistant Professor, Department of Computer Science. After that, the meeting was formally started by the Dean, Faculty of Science by briefing the members about the agenda items.

The following decisions and recommendations were taken during the meeting along with significant points of discussion.

Agenda 01. Confirmation of minutes of 7th BoS

Background:

The minutes of the 7th meeting of the Board of Study (BoS) was presented to the forum for information.

Discussion and Decision:

The forum endorsed the minutes.

Agenda 02. Course codes for BS in Computer Science Program

Background:


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The course codes were assigned to the courses during the 7th meeting of the Board of Studies and the same were sent to the Board of faculty and Academic council. However, the Academic council send it back with minor modifications in domain subjects to the BoS.

Discussion and Decision:

The BoS recommends following revised course coding and scheme of studies as per the University and HEC guidelines of the undergraduate policy 2023 respectively.

2.1 Scheme of study for BS Computer Science

Semester	Course Code	Subject	Cr. Hrs.
1st	CS-111	Application of Information & Communication Technologies	3(2-3)
	CS-121	Programming Fundamentals	4(3-3)
	ENG-113	Functional English	3(3-0)
	ISL-111	Islamic Studies	2(2-0)
	-	Civics and Community Engagement	2(2-0)
	CS-101	Mathematics-I (for Pre-Medical students)	
		Applied physics	3(2-3)
		Total	17(14-9)
2nd	-	Linear Algebra	3(3-0)
	ENG-213	Technical and Business Writing	3(3-0)
		Ideology and Constitution of Pakistan	2(2-0)
	CS-131	Digital Logic Design	3(2-3)
	CS-122	Object Oriented Programming	4(3-3)
		Entrepreneurship	2(2-0)
	-	Total	17(15-6)
3rd	CS-221	Data Structure	4(3-3)
	STAT-231	Probability and Statistics	3(3-0)
	CS-231	Human Computer Interaction & Computer Graphics	3(2-3)
		Introduction to Management	2(2-0)
		Expository Writing	3(3-0)
	CS-222	Web Technologies (Domain Elective 1)	3(2-3)
	-	Total	18(15-9)

4th	CS-223	Compiler Construction	3(2-3)
	CS-232	Analysis of Algorithms	3(3-0)
	-	Introduction to Marketing (Social Science)	3(3-0)
		Calculus and Analytical Geometry	3(3-0)
		Web Engineering (Domain Elective 5)	3(3-0)
	-	Total	15(14-3)
5th	CS-331	Operating Systems	3(2-3)
	CS-321	Software Engineering	3(3-0)
	CS-332	Theory of Automata	3(3-0)
	CS-333	Computer Architecture	3(2-3)
	CS-341	Data Communication and Computer Networks	4(3-3)
	-	Total	16(13-9)
6th	CS-334	Computer Organization & Assembly Language	3(2-3)
	CS-351	Database Systems	4(3-3)
	CS-361	Artificial Intelligence	3(2-3)
	CS-322	Advance Programming – Visual Pro	3(2-3)
	CS-323	Mobile App Development	3(2-3)
	-	Total	16(11-15)
7th	CS-421	Software Testing & Quality Assurance (Domain Elective)	3(2-3)
	-	Professional Practices	2 (2-0)
	CS-431	Parallel & Distributed Computing	3(2-3)
	CS-441	Information Security	3(2-3)
	-	Multivariable Calculus	3(3-0)
	CS-598	Final Year Project – I	2(0-6)
	-	Total	16(11,15)
8th	-	Numerical Analysis	3(2-3)
	CS-432	Discrete Structures	3(3-0)
	CS-442	Cyber Security	3(2-3)
	CS-452	Advance database management system	3 (2-3)
	CS-599	Final Year Project – II	4(0-12)
	-	Total	16(9-21)


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2.2 Course codes and scheme of study of MS Computer Science

Background:

After thorough discussion, the BoS recommended the following course codes and scheme of studies for MS Computer Science.

S. No	Domain	Code	Course Code	Credit Hours	Course Title
1	System Design	1	CS611	3(3,0)	Advanced Analysis of Algorithms
2			CS612	3(3,0)	Advanced Operating Systems
3			CS613	3(3,0)	Theory of Automata – II
4			CS614	3(3,0)	Advanced Computer Architecture
5	Data Base Systems	2	CS721	3(3,0)	Statistical and Mathematical Methods for Data Science
6			CS722	3(2,1)	Tools and Techniques in Data Science
7			CS723	3(3,0)	Advanced Data Mining I
8	Artificial Intelligence	3	CS731	3(3,0)	Machine Learning
9			CS732	3(3,0)	Natural Language Processing I
10			CS733	3(3,0)	Soft Computing
11	Software Engineering	4	CS641	3(3,0)	Requirements Engineering
12			CS642	3(3,0)	Component Based Software Engineering
13			CS643	3(3,0)	Advanced Software System Architecture I
14			CS741	3(3,0)	Agile Software Development
15			CS742	3(3,0)	Software Metrics
16			CS743	3(3,0)	Software Testing and Quality Assurance
17			CS744	3(3,0)	Formal Methods
18			CS745	3(3,0)	Advanced Human-Computer Interaction I
19			CS746	3(3,0)	Advanced Software Project Management I
20			CS747	3(3,0)	Software Configuration Management
21			CS748	3(3,0)	Software Process Management
22			CS651	3(3,0)	Advanced Computer Networks I
23			CS652	3(3,0)	Complex Networks
24			CS751	3(3,0)	Wireless Networks

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25	Computer Networks	5	CS752	3(3,0)	Wireless Sensor Networks
26			CS753	3(3,0)	Distributed Systems
27			CS754	3(3,0)	Distributed Data Processing
28			CS755	3(3,0)	Networks Security
29			CS756	3(3,0)	Information Privacy and Security
30			CS757	3(3,0)	Securing the Internet of Things
31			CS758	3(3,0)	Topics in Applied Cryptography
32	Research	9	CS690	3(3,0)	Research Methodology I
33			CS798	3(3,0)	MS Thesis (Part I)
34			CS799	3(3,0)	MS Thesis (Part II)

Scheme of studies for MS Computer Science

	Course Code	Credit Hours	Course Title	Comment ¹
Semester 1	CS611	3(3,0)	Advanced Analysis of Algorithms	
	CS612	3(3,0)	Advanced Operating Systems	
	CSxyz		Elective -I	
	CSxyz		Elective -II	
Semester 2	CS613	3(3,0)	Theory of Automata – II	
	CS614	3(3,0)	Advanced Computer Architecture	
	CSxyz		Elective -I	
	CSxyz		Elective -II	
Semester 3	CS798	3(0,3)	MS Thesis (Part I)	Research
Semester 4	CS799	3(3,0)	MS Thesis (Part II)	Write-up

2.3 Scheme of study for PhD Computer Science

The BoS recommended the following course codes and scheme of studies for PhD Computer Science for further approval.

S. No	Domain	Code	Course Code	Credit Hours	Course Title
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Research

3	System Design		CS813	3(3,0)	Advanced Theory of Automata
5	Data Base Systems	2	CS821	3(3,0)	Advanced Statistical and Mathematical Methods for Data Science
6			CS822	3(2,1)	Advanced Tools and Techniques in Data Science
7			CS823	3(3,0)	Advanced Data Mining II
8	Artificial Intelligence	3	CS831	3(3,0)	Advanced Machine Learning
9			CS832	3(3,0)	Natural Language Processing II
10			CS833	3(3,0)	Advanced Soft Computing
11	Software Engineering	4	CS841	3(3,0)	Advanced Requirements Engineering
12			CS842	3(3,0)	Component Based Software Engineering
13			CS843	3(3,0)	Advanced Software System Architecture
14			CS841	3(3,0)	Agile Software Development
15			CS842	3(3,0)	Software Metrics
16			CS843	3(3,0)	Software Testing and Quality Assurance
17			CS844	3(3,0)	Advanced Formal Methods
18			CS845	3(3,0)	Advanced Human-Computer Interaction II
19			CS846	3(3,0)	Advanced Software Project Management II
20			CS847	3(3,0)	Advanced Software Configuration Management
21			CS848	3(3,0)	Advanced Software Process Management
22	Computer Networks	5	CS851	3(3,0)	Advanced Computer Networks II
23			CS852	3(3,0)	Advanced Complex Networks
24			CS851	3(3,0)	Advanced Wireless Networks
25			CS852	3(3,0)	Advanced Wireless Sensor Networks
26			CS853	3(3,0)	Advanced Distributed Systems
27			CS854	3(3,0)	Advanced Distributed Data Processing
28			CS855	3(3,0)	Advanced Networks Security
29			CS856	3(3,0)	Advanced Information Privacy and Security
30			CS857	3(3,0)	Advanced Securing the Internet of Things
31			CS858	3(3,0)	Advanced Topics in Applied Cryptography
32			CS890	3(3,0)	Research Methodology II
35			CS996	3(3,0)	Ph.D. Dissertation (Part I)



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36	Research	9	CS997	3(3,0)	Ph.D. Dissertation (Part II)
37			CS998	3(3,0)	Ph.D. Dissertation (Part III)
38			CS999	3(3,0)	Ph.D. Dissertation (Part IV)

	Semester 1			
	Course Code	Credit Hours	Course Title	Comments
Semester 1	CSxyz	3(3,0)	Elective -I	
	CSxyz	3(3,0)	Elective -II	
	CSxyz	3(3,0)	Elective -III	
Semester 2	CSxyz	3(3,0)	Elective -IV	
	CSxyz	3(3,0)	Elective -V	
	CSxyz	3(3,0)	Elective -VI	
Semester 3	CS996	3(0,3)	Ph.D. Dissertation (Part I)	Research
Semester 4	CS997	3(0,3)	Ph.D. Dissertation (Part II)	Research
Semester 5	CS998	3(0,3)	Ph.D. Dissertation (Part III)	Research
Semester 6	CS999	3(3,0)	Ph.D. Dissertation (Part IV)	Write-up

2.4 Course codes for subjects offered by the other departments of the university

Course codes and syllabus for the subjects that are requested from other departments of the university will be provided by their respective departments approved from their respective Board of Studies.

Discussion and Decision:

The forum recommended for approval the course coding for MS and PhD programmes.

Agenda 03. Starting of new programs in BS

Background:

The Department of Computer Science receives over 900 applications for admission to the BS in Computer Science program, out of which only 200 students are securing admission. The remaining students secure admission in other universities. Considering the growing demand for AI

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professionals, data science, and cybersecurity in various organizations and industries, initiating these programs will equip students with valuable skills, aligning with current industry trends and creating new career opportunities. The following BS programs were presented to the forum:

- BS in Artificial Intelligence (Annexed I)
- BS in Data Science (Annexed II)
- BS in Cyber Security (Annexed III)

Discussion and Decision:

Keeping in view the existing high number of applicants and market demand, the forum strongly recommended starting of the approved and aforementioned new programmes. Furthermore, the forum emphasized to fulfill the requirements of the Council for existing and new programmes.

Any other agenda

a. Admission in 5th Semester based on the Associate Degree program

Background:

The Department of Computer Science received applications for enrollment in 5th semester based on the four-semester associated degree as per the advertisement by the university for fall- 2023. The applicants have various deficiencies such mismatch of subjects' order, credit hours, CGPA, and so on.

Discussion and Decision:

The forum recommended to forward the case to the Director Academics for guidance and final approval.

b. Course codes for the computer subjects offered in Department of Management Sciences,

Background:

The course coding for outer department of Management were discussed.

Discussion and Decision:

The forum recommended the following course codes for the subject below.

1. Course code for "Introduction to Computer" is CS-114
2. Course code for "Office Management Tools" is CS-112
3. Course code for "Management Information System" is CS-113

The meeting concluded with vote of thanks by the convener.

